

Make 2 two tables Employee and Department.

```
CREATE DATABASE internship_db;
```

```
USE internship_db;
```

//Department Table

```
CREATE TABLE Department (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50)  
);
```

//Employee Table

```
CREATE TABLE Employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    salary DECIMAL(10,2),  
    hire_date DATE,  
    dept_id INT,  
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id)  
);
```

// Values for department table

```
INSERT INTO Department VALUES  
(1,'HR'),  
(2,'IT'),  
(3,'Finance');
```

// Values for Employee

INSERT INTO Employee VALUES

```
(101,'Pranali',50000,'2025-10-15',1),
(102,'Kajal',70000,'2025-11-20',2),
(103,'Amit',60000,'2025-12-05',2),
(104,'Sneha',80000,'2026-02-10',3),
(105,'Karan',55000,'2026-01-15',1),
(106,'Neha',90000,'2026-06-01',2),
(107,'Vikas',65000,'2026-03-12',3),
(108,'Pooja',72000,'2026-04-25',1),
(109,'Rohan',85000,'2026-05-18',2),
(110,'Anjali',58000,'2026-02-28',3),
(111,'Pranali',50000,'2025-09-10',1),
(112,'Kajal',70000,'2025-08-25',2);
```

//To view Deapartment Table

SELECT * FROM Department;

37 • SELECT * FROM Department;
38
39 • SELECT * FROM Employee;
40

Result Grid |  Filter Rows: | Edit:    | Export/Import:

	dept_id	dept_name
▶	1	HR
	2	IT
	3	Finance
★	NULL	NULL

//To view Employee Table

```
SELECT * FROM Employee;
```

```
39 • SELECT * FROM Employee;
40
```

emp_id	emp_name	salary	hire_date	dept_id
102	Kajal	70000.00	2025-11-20	2
103	Amit	60000.00	2025-12-05	2
104	Sneha	80000.00	2026-02-10	3
105	Karan	55000.00	2026-01-15	1
106	Neha	90000.00	2026-06-01	2
107	Vikas	65000.00	2026-03-12	3
108	Pooja	72000.00	2026-04-25	1
109	Rohan	85000.00	2026-05-18	2
110	Anjali	58000.00	2026-02-28	3
111	Pranali	50000.00	2025-09-10	1
112	Kajal	70000.00	2025-08-25	2
•	NULL	NULL	NULL	NULL

Question 1: Top 5 Highest Salary Employees

```
SELECT *
FROM Employee
ORDER BY salary DESC
LIMIT 5;
```

```
41 -- Question 1: Top 5 Highest Salary Employees
42 • SELECT *
43 FROM Employee
44 ORDER BY salary DESC
45 LIMIT 5;
46
```

emp_id	emp_name	salary	hire_date	dept_id
106	Neha	90000.00	2026-06-01	2
109	Rohan	85000.00	2026-05-18	2
104	Sneha	80000.00	2026-02-10	3
108	Pooja	72000.00	2026-04-25	1
102	Kajal	70000.00	2025-11-20	2
•	NULL	NULL	NULL	NULL

Question 2: Department Wise Employee Count

```
SELECT dept_id, COUNT(*) AS employee_count  
FROM Employee  
GROUP BY dept_id;
```

```
47      -- Question 2: Department Wise Employee Count  
48 •    SELECT dept_id, COUNT(*) AS employee_count  
49      FROM Employee  
50      GROUP BY dept_id;  
51  
52
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	dept_id	employee_count			
▶	1	4			
	2	5			
	3	3			

Question 3: Second Highest Salary

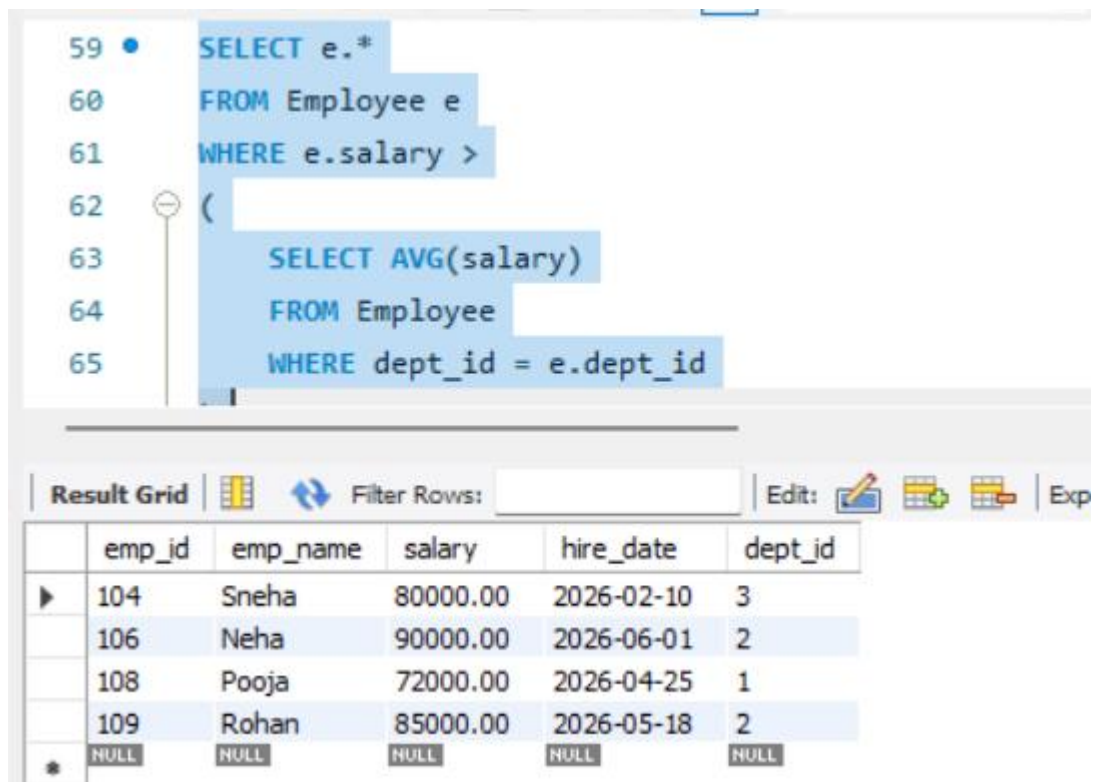
```
SELECT MAX(salary) AS second_highest_salary  
FROM Employee  
WHERE salary < (SELECT MAX(salary) FROM Employee);
```

```
--  
53      -- Question 3: Second Highest Salary  
54 •    SELECT MAX(salary) AS second_highest_salary  
55      FROM Employee  
56      WHERE salary < (SELECT MAX(salary) FROM Employee);  
57
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	second_highest_salary			
▶	85000.00			

Q4. Employees whose salary > department average salary

```
SELECT e.*
FROM Employee e
WHERE e.salary >
(
  SELECT AVG(salary)
  FROM Employee
  WHERE dept_id = e.dept_id
);
```



The screenshot shows a SQL IDE interface. The top pane displays the SQL query for finding employees whose salary is greater than the department average. The bottom pane shows the 'Result Grid' with 5 columns: emp_id, emp_name, salary, hire_date, and dept_id. The results include 5 rows: employee 104 (Sneha, salary 80000.00, dept 3), employee 106 (Neha, salary 90000.00, dept 2), employee 108 (Pooja, salary 72000.00, dept 1), employee 109 (Rohan, salary 85000.00, dept 2), and a row with all NULL values.

	emp_id	emp_name	salary	hire_date	dept_id
▶	104	Sneha	80000.00	2026-02-10	3
	106	Neha	90000.00	2026-06-01	2
	108	Pooja	72000.00	2026-04-25	1
	109	Rohan	85000.00	2026-05-18	2
*	NULL	NULL	NULL	NULL	NULL

Q5. Inner Join

```
SELECT e.emp_id,  
       e.emp_name,  
       d.dept_name  
FROM Employee e  
INNER JOIN Department d  
ON e.dept_id = d.dept_id;
```

```
69 • SELECT e.emp_id,  
70       e.emp_name,  
71       d.dept_name  
72 FROM Employee e  
73 INNER JOIN Department d  
74 ON e.dept_id = d.dept_id;  
75
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content

	emp_id	emp_name	dept_name
▶	101	Pranali	HR
	105	Karan	HR
	108	Pooja	HR
	111	Pranali	HR
	102	Kajal	IT
	103	Amit	IT
	106	Neha	IT
	109	Rohan	IT
	112	Kajal	IT
	104	Sneha	Finance
	107	Vikas	Finance
	110	Anjali	Finance

Q6. Left Join

```
SELECT e.emp_id,  
       e.emp_name,  
       d.dept_name  
FROM Employee e  
LEFT JOIN Department d  
ON e.dept_id = d.dept_id;
```

```
76 -- Q6. Left Join  
77 • SELECT e.emp_id,  
78       e.emp_name,  
79       d.dept_name  
80 FROM Employee e  
81 LEFT JOIN Department d  
82 ON e.dept_id = d.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	emp_id	emp_name	dept_name
▶	101	Pranali	HR
	102	Kajal	IT
	103	Amit	IT
	104	Sneha	Finance
	105	Karan	HR
	106	Neha	IT
	107	Vikas	Finance
	108	Pooja	HR
	109	Rohan	IT
	110	Anjali	Finance
	111	Pranali	HR
	112	Kajal	IT

Q7. Group By with Having

```
SELECT dept_id,  
       COUNT(*) AS total_employees  
FROM Employee  
GROUP BY dept_id  
HAVING COUNT(*) > 3;
```

```
84      -- Q7. Group By with Having  
85 •    SELECT dept_id,  
86           COUNT(*) AS total_employees  
87      FROM Employee  
88      GROUP BY dept_id  
89      HAVING COUNT(*) > 3;
```

Result Grid | Filter Rows: | Export: | Wrap

	dept_id	total_employees
▶	1	4
	2	5

Q8. Employees hired in last 6 months

```
SELECT *  
FROM Employee  
WHERE hire_date >= DATE_SUB(CURDATE(), INTERVAL 6 MONTH);
```

```
91      -- Q8. Employees hired in last 6 months  
92 •    SELECT *  
93      FROM Employee  
94      WHERE hire_date >= DATE_SUB(CURDATE(), INTERVAL 6 MONTH);  
95
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

	emp_id	emp_name	salary	hire_date	dept_id
▶	104	Sneha	80000.00	2026-02-10	3
	105	Karan	55000.00	2026-01-15	1
	106	Neha	90000.00	2026-06-01	2
	107	Vikas	65000.00	2026-03-12	3
	108	Pooja	72000.00	2026-04-25	1
	109	Rohan	85000.00	2026-05-18	2
	110	Anjali	58000.00	2026-02-28	3
*	NULL	NULL	NULL	NULL	NULL

Q9. Find Duplicate Records

```
SELECT emp_name,  
       COUNT(*)  
FROM Employee  
GROUP BY emp_name  
HAVING COUNT(*) > 1;
```

```
135 -- Q9. Find Duplicate Records  
136 • SELECT emp_name,  
137       COUNT(*)  
138 FROM Employee  
139 GROUP BY emp_name  
140 HAVING COUNT(*) > 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell

	emp_name	COUNT(*)
▶	Pranali	2
	Kajal	2

Q10. How to Remove Duplicate Records

```
DELETE e1  
FROM Employee e1  
JOIN Employee e2  
ON e1.emp_name = e2.emp_name  
AND e1.emp_id > e2.emp_id;
```

```
39 • SELECT * FROM Employee;  
40
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	emp_id	emp_name	salary	hire_date	dept_id
▶	101	Pranali	50000.00	2025-10-15	1
	102	Kajal	70000.00	2025-11-20	2
	103	Amit	60000.00	2025-12-05	2
	104	Sneha	80000.00	2026-02-10	3
	105	Karan	55000.00	2026-01-15	1
	106	Neha	90000.00	2026-06-01	2
	107	Vikas	65000.00	2026-03-12	3
	108	Pooja	72000.00	2026-04-25	1
	109	Rohan	85000.00	2026-05-18	2
	110	Anjali	58000.00	2026-02-28	3